JAN-P-156A

9 JULY 1946

SUPERSEDING
JAN–P–156
30 NOVEMBER 1944

JOINT ARMY-NAVY SPECIFICATIONS

POTASSIUM NITRATE

Army Number	Navy Number
50-11-26D	51P31a

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
50–11–26C	51P31
30 Nov. 1944	30 Nov. 1944

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS.

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms Ammunition.[1]

100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard. Testing.

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

INTERSTATE COMMERCE COMMISSION PUBLICATION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

B. GRADES AND CLASSES.

B–1. This specification covers one grade of potassium nitrate

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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furnished in the following classes as specified in the contract or order (see par. H–1):

Class a
Class b
Class c

C. MATERIAL AND WORKMANSHIP.

C–1. See sections D and E.

D. GENERAL REQUIREMENTS.

D–1. Potassium nitrate shall conform to the chemical requirements shown in table I:

TABLE I—*Chemical requirements*

	Percent
Moisture (max.)	0.2
Acidity or alkalinity	None
Insoluble material (max.)	0.1
Grit	None
Chlorides as KCl (max.)	0.07
Chlorates and perchlorates as $KClO_4$ (max.)	0.5
Iron and aluminum as oxides (max.)	0.5
Calcium and magnesium as oxides (max.)	0.5
Sodium as oxide (max.)	0.25
Ammonium compounds	None
Nitrogen (min.)	13.77

D–2. *Color.*—The color shall be white. An aqueous solution (1 to 25) shall be free from any color and show no opalescence.

E. DETAIL REQUIREMENTS.

E–1. *Granulation.*—

E–1a. Class a potassium nitrate shall be furnished in lump form.

E–1b. Class b and c potassium nitrate shall conform to the granulation requirements shown in table II, using U. S. Standard sieves conforming to the requirements of Federal Specification RR–S–366.

TABLE II—*Granulation*

Through sieve No.	*Percent minimum*	
	class b	class c
60	99.9	—
80	—	99.9
100	40	—

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Size of lot.*—For purposes of inspection, a lot shall not exceed 10,000 pounds and shall consist of material from not more than one manufacturing batch of potassium nitrate (see par. H–5).

F–2. *Sampling.*—Three containers shall be selected at random

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from the lot for sampling purposes. If there are fewer than 3 containers in the lot, all the containers shall be sampled. A 2-pound representative sample shall be taken from each container so chosen. Each sample shall be placed in a clean, moisture-free, air-tight container. Each container shall be sealed and labeled to identify the sample with the lot represented. Each sample shall be subjected to all the tests specified in paragraph F–4.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made in accordance with the requirements of U. S. Army Specification 50–0–1.

F–3b. *Navy.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—The laboratory tests shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory, unless otherwise specified in the contract or order.

F–4a. *Moisture.*—Dry an accurately weighed sample of approximately 10 gm in a tared dish at 150° C. to constant weight; cool in a desiccator and weigh. Calculate the loss in weight as percentage of moisture.

F–4b. *Acidity or alkalinity.*—Dissolve a weighed portion of approximately 1 gm of the sample in 20 ml of neutral distilled water and determine if sensitive blue or red litmus paper is changed in color within 3 minutes when immersed in the solution. Blue litmus paper shall be considered to be sufficiently sensitive if it shows a distinct reddish coloration after being immersed for 3 minutes in N/1000 hydrochloric or sulfuric acid. Similarly, red litmus paper should be considered sensitive enough if it shows a distinct bluish coloration, after being immersed for 3 minutes in N/1000 sodium hydroxide.

F–4c. *Insoluble material.*—Dissolve a weighed portion of approximately 10 gm of the sample in 100 ml of boiling distilled water and filter the solution through a tared filtering crucible. Wash the insoluble residue on the filter with hot water. Dry the crucible and contents at 100° to 105° C. for 3 hours, cool in a desiccator, and weigh. Calculate the increase in weight as percentage of insoluble material.

F–4d. *Grit.*—Dissolve a 25-gm portion of the sample in water, filter and wash thoroughly. Transfer any material retained on the filter to a smooth glass slide and press and rub the material with a smooth steel spatula. The presence of grit is indicated by the persistence of a scratching noise.

F–4e. *Chlorides.*—Dissolve an accurately weighed portion of approximately 5 gm of the sample in 100 ml of distilled water and make the solution slightly acid with nitric acid. Add a few drops of approximately 0.25 N silver nitrate solution. If more than a faint opalescence results, add more silver nitrate solution until precipitation is complete. Boil for 2 minutes with constant stirring, and allow the precipitate to settle overnight in a dark place. Decant the solution through a tared filtering crucible. Wash the precipitate 4 times by decantation, using 20-ml portions of water

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containing 1 ml of concentrated nitric acid and 2 drops of silver nitrate solution per 100 ml. Transfer the precipitate to the crucible with 20 ml of the same wash water, wash it with 20 ml of 1-percent nitric acid solution and with two 20-ml portions of distilled water. Dry the crucible and precipitate for 4 hours at 135° C., cool in a desiccator, and weigh. Calculate the percentage of chlorides as potassium chloride as follows:

$$\text{Percent potassium chloride} = \frac{52A}{W}$$

where A = weight of precipitate
W = weight of sample.

F–4f. *Chlorates and perchlorates.*—Transfer an accurately weighed portion of approximately 2 gm of the sample, to a platinum crucible, and mix with approximately 10 gm of anhydrous sodium carbonate. Cover the mixture with a layer of sodium carbonate, fuse, and cool. Dissolve the fused mass in distilled water, render the solution slightly acid with nitric acid, and proceed with the precipitation as described in paragraph F–4e. Calculate the percentage of total chlorates and perchlorates as potassium perchlorate as follows:

$$\text{Percent potassium perchlorate} = \frac{97(A\text{-}0.019BW)}{W}$$

where A = weight of precipitate
B = percent potassium chloride as determined in paragraph F–4e
W = weight of sample.

F–4g. *Iron and aluminum.*—Dissolve a weighed portion of approximately 5 gm of the sample in distilled water and filter the solution. Acidify the filtrate with nitric acid, add a slight excess of ammonium hydroxide, and boil. Catch the precipitate on an ashless filter and wash with hot distilled water. Retain the filtrate and washings for further analysis. Ignite the filter paper and precipitate in a tared porcelain crucible to constant weight. Cool the crucible and contents in a desiccator and weigh. Calculate the increase in weight as percentage of iron and aluminum oxides.

F–4h. *Calcium and magnesium.*—Dilute the combined filtrate and washings from the determination of iron and aluminum (see par. F–4g) to about 200 ml and boil. Evaporate the filtrate to a volume of about 20 ml. Filter, and add 15 ml of 9N ammonium carbonate solution and 15 ml of 95-percent ethyl alcohol. Allow the mixture to stand for at least 30 minutes, with frequent stirring. Filter through paper and wash the precipitate with 9N ammonium carbonate solution. Dissolve the precipitate in dilute hydrochloric acid, dilute to about 50 ml, boil, and add 1 ml of saturated ammonium oxalate solution. Allow the solution to stand for 12 hours, filter through a tared filtering crucible, and wash the precipitate with hot water containing ammonium oxalate. Ignite the precipitate, gently at first and then with a blast lamp, cool the crucible and contents in a desiccator, and weigh. Calculate the increase in weight as percentage of calcium oxide. Concentrate the filtrate from the calcium determination to about 50

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ml. Add 10 ml of a 10-percent solution of ammonium phosphate, and 20 ml of 10-percent solution of ammonia. Allow the solution to stand for 4 or 5 hours, filter through a tared filtering crucible and wash with a 2.5-percent solution of ammonia. Dry the precipitate and ignite to constant weight with a blast lamp. Cool the crucible and contents in a desiccator, and weigh. Calculate the increase in weight to percentage of magnesium oxide as follows:

$$\text{Percent magnesium oxide} = \frac{36A}{W}$$

where A = weight of precipitate
W = weight of sample.

F–4i. *Sodium.*—

F–4i(1). *Magnesium uranyl acetate reagent.*—Prepare solution A by dissolving 90 gm of uranyl acetate, $UO_2(C_2H_3O_2)_2 2H_2O$, and 60 ml of glacial acetic acid in sufficient distilled water to make a volume of 1 liter by heating to 70° C, and stirring until solution is complete. Prepare solution B by dissolving 600 gm of magnesium acetate, $Mg(C_2H_3O_2)_2 4H_2O$, in 60 ml of glacial acetic acid and sufficient distilled water to make a volume of 1 liter by heating to 70 °C. and stirring until solution is complete. Mix together solutions A and B while at 70° C., and cool the mixture to 20° C. After allowing the mixture to stand at this temperature for 2 hours or longer, filter it through a dry filter paper into an amber-colored bottle. Store the bottle containing the solution where it is not exposed to direct sunlight. If a precipitate appears on standing, filter the solution again prior to use.

F–4i(2). *Alcohol wash liquid.*—Prepare a wash liquid as follows: Shake 1 ml of 1-percent sodium chloride solution with 25 ml of magnesium uranyl acetate reagent. Catch the precipitate on a filter paper and wash thoroughly with 95-percent ethyl alcohol. Mix the precipitate with 1 liter of 95-percent ethyl alcohol and allow to stand for 1 hour with frequent shaking. Filter the solution and use the filtrate as a wash liquid.

F–4i(3). *Procedure.*—Transfer a weighed portion of approximately 0.5 gm of the sample to a 150-ml beaker and dissolve in 5 ml of distilled water and cool to 20° C. Add 25 ml of the magnesium uranyl acetate reagent (see par. F–4i(1)) which has previously been cooled at 20° C. While maintaining the temperature at 20° C., rapidly stir the contents of the beaker for ½ hour by means of a mechanical stirrer. Transfer the precipitate to a tared filtering crucible, using a total of 25 to 30 ml of the wash liquid, specified in par. F–4i(2), for transferring and washing the precipitate. Dry the crucible and precipitate at 100° to 110° C. for ½ hour, cool in a desiccator, and weigh. Calculate the weight of the magnesium sodium uranyl acetate,

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$MgNa(UO_2)_3(C_2H_3O_2)_9{:}6\frac{1}{2}\ H_2O$ to percentage of sodium oxide in the sample.

$$\text{Percent sodium oxide} = \frac{2.06A}{W}$$

where A = weight of precipitate
W = weight of sample.

F–4j. *Ammonium compounds.*—Transfer an accurately weighed portion of approximately 1.0 gm of the sample to a 150-ml beaker, dissolve in 25 ml of distilled water, and add sufficient excess of concentrated sodium hydroxide solution to render the potassium nitrate solution alkaline. Heat the mixture and test the vapors with a moist strip of red litmus paper of such sensitivity that it changes color in 30 seconds when immersed in 0.0005 N alkali and stirred continuously. Care should be taken that no spray from the solution touches the test paper. Note if one or more ammonium compounds are present in the sample as indicated by the paper becoming blue in color.

F–4k. *Nitrogen.*—

F–4k(1). *Standardization of the nitrometer.*—

F–4k(1)a. *Absolute method.*—Assemble the nitrometer (du Pont or equivalent) after thoroughly cleaning and drying all parts. The various parts, with the exception of the reservoir bulbs, are completely filled with mercury. Place 25 ml of 94.5 ± 0.5-percent sulfuric acid in the generating bulb, draw air into the bulb to nearly the level of the lower shoulder by lowering the mercury reservoir and opening both stopcocks. Close both stopcocks and shake vigorously for 3 minutes. Connect the generating bulb into the compensating tube by adjustment of the 2 mercury reservoirs, keeping open the stopcock of the measuring tube. Continue the introduction of the air until the mercury in the reading tube is nearly at the 12.50 mark. Seal the compensating tube by means of a small blowpipe flame. Desiccate similarly in the generating bulb another portion of air and transfer this to the reading tube until the mercury in the reading tube, compensating tube, and their mercury reservoirs is at the same level, the mercury in the reading tube being approximately at the 12.50 mark. It is not necessary that these gas volumes be exact or equal. Fill a small U-tube manometer with sulfuric acid and attach to the reading tube. Allow the apparatus to come to constant temperature and then adjust the mercury levels in the reading tube, compensating tube, and their mercury reservoirs so that the air in the reading tube is under atmospheric pressure, as shown by the acid levels in the arms of the manometer tube. Take the reading on the graduated scale of the calibrated reading tube, and at the same time note the temperature and barometric pressure. From these readings calculate the volume of air at 20° C. and 760 mm pressure in the reading tube by means of the following formula:

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$$V = R\frac{P \times 293\ (1-0.0018t)}{760(273+T)\ (1-0.00018 \times 20)} = \frac{0.3869\ PR(1-0.00018t)}{273+T}$$

where V = ml air at 20° C. and 760 mm, barometer at 20° C.
R = ml air at the observed temperature (70° C.) and at the observed pressure (Pmm)
t = temperature (° C.) of the barometer.

The factor 0.00018 represents the correction to be applied for the coefficient of expansion of mercury in the barometer. When V has been calculated, adjust the heights of the reading tube, compensating tube, and their reservoirs so that, with the mercury in the reading and compensating tubes on a level, the mercury in the reading tube marks the calculated volume of the air. Paste a strip of paper on the tube so as to mark the height of the mercury in the compensating tube. Take this as the standard volume of air with which every volume of gas to be measured is compared.

Note: The use of dry air tends to form a scum around the reading level of the compensating tube, thus requiring frequent restandardization of the nitrometer. This scum is caused by reaction of oxygen in the air with mercury or by impurities. Dry nitrogen eliminates this condition and should be used, if avail- A half drop of sulfuric acid will also prevent this scum.

F–4k(1)b. *Potassium nitrate method.*—Recrystallize three times from 95–percent alcohol a sample of reagent grade potassium nitrate, and grind this to pass through a U. S. Standard No. 100 sieve. After drying at 135° to 150° C. for 2 or 3 hours, weigh a portion of exactly 1 gm into a small weighing bottle and transfer this portion to the cup of the generating bulb. Wash the potassium nitrate into the generating bulb by means of 8 to 10 portions of 94.5 ± 0.5-percent nitrogen-free sulfuric acid cooled to approximately 5° C., using a total volume of 25 ml of the acid. With the lower stopcock of the generating bulb open and the mercury reservoir low enough to maintain a slightly reduced pressure in the generating bulb, shake the sample until most of the gas has been generated. Adjust the mercury reservoir until the mercury drops nearly to the lower shoulder of the generating bulb. Close the lower stopcock and shake the bulb vigorously for 3 minutes. Replace the bulb on the rack, open the lower stopcock, and allow the tube to stand for 5 minutes, the levels in the generating bulb and reservoir being approximately the same.[1] Close the lower stopcock, and shake the generating bulb for 1 additional minute. Transfer the gas to the reading tube, and adjust the levels of the mercury in the reading and compensating tubes to approximately the same height. Allow the gas to stand for about 20 minutes to permit equalization of the temperature. With a sulfuric acid U-tube leveling device, make careful adjustment, so that the mercury in both tubes is at the same level and that in the reading tube is at the 13.85 mark (the percentage of nitrogen in potassium nitrate). Make a mark on the compensating tube to show the volume occupied by the air equivalent to the volume of nitric oxide gas liberated from

[1] Having previously introduced desiccated air into the Compensating tube as described under the Absolute Method.

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exactly 1 gm of potassium nitrate. This mark is usually made on a strip of gummed paper attached to the compensating tube.

Note 1.—In order to avoid any doubt regarding the purity of the potassium nitrate, it is desirable to check the standardization obtained by the use of potassium nitrate by means of the absolute method. If the agreement is within one or two hundredths of one percent, the potassium nitrate can be considered as of satisfactory quality and used in subsequent periodic standardizations. It has been found by actual experimentation that the solubility of nitric oxide in sulfuric acid of the strength prescribed is so small that no correction need be applied to bring the value in agreement with the mark set by the absolute method.

Note 2.—If the nitrometer is standardized by both the absolute and potassium nitrate methods, the results may not be in agreement by 0.01 to 0.04-percent nitrogen due to expansion of the brass barometer scale. Such difference may be removed by applying the brass scale correction corresponding to the temperature (see par. H–3.)

F–4k(2). *Procedure.*—Crush a portion of about 3 gm of the sample so that it will pass through a U. S. Standard No. 100 sieve and dry for 2 hours at 150°C. Accurately weigh approximately 1.0 gm of the dried sample and transfer it to the cup of a standardized 5–part nitrometer (du Pont or equivalent). Measure into a small graduate 25 ml of nitrogen-free 94.5 ± 0.5-percent sulfuric acid cooled to approximately 5°C. Wash out the weighing bottle into the nitrometer cup with 5 ml of the sulfuric acid. Lower the mercury reservoir sufficiently to maintain a reduced pressure in the generating bulb. Open wide the upper stopcock and open the lower stopcock sufficiently to draw the acid mixture into the generating bulb. Make successive 4-ml rinsings with the rest of the sulfuric acid until it has all been drawn in. Close the upper stopcock, leaving the lower one open, and adjust the mercury reservoir just low enough to maintain a slightly reduced pressure in the generating bulb. Shake the bulb gently until most of the gas has been generated. Lower the mercury reservoir until the level of the mercury in the generating bulb is at the height of the lower shoulder, then close the lower stopcock and shake vigorously for 3 minutes. Replace the bulb on the rack, open the lower stopcock, and adjust the mercury reservoir until the mercury in the generating bulb is at approximately the same height as the mercury in the reservoir. This will bring the pressure inside the generating bulb to approximately one atmosphere, and the solubility of the nitric oxide gas in the sulfuric acid will be normal. Close the lower stopcock and shake the bulb vigorously for an additional 1 minute. Transfer the gas to the measuring tube, adjust the levels of the mercury in the compensating and measuring tubes to approximately the same height. Let the gas stand for 20 minutes in order to permit equalization of the temperature of the gas in the two tubes, and adjust the mercury levels closely with a sulfuric acid U-tube leveling device, the mercury level in the

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compensating tube being at the standard volume mark. Calculate the percentage of nitrogen as follows:

$$\text{Percent nitrogen} = \frac{R}{W}$$

where R = reading of the measuring tube
W = weight of sample.

F–4l. *Color.*—Dissolve 5 gm of the sample in 125 ml of distilled water. Note if the resulting solution is clear and free from any color or opalescence.

F–4m. *Granulation.*—Nest the specified sieve or sieves on a bottom pan. Place a weighed portion of 100 gm of the sample on the upper sieve, cover this, and shake for 10 minutes by hand or 5 minutes by means of a mechancial shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amounts retained or passing through the sieves, and calculate to percentages as required.

F–5. *Rejection and resubmission.*—If any of the samples does not conform to any of the specified requirements, the entire lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on each container at no expense to the Government. The contractor may then remove defective portions of the lot and resubmit the lot for acceptance.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—Unless otherwise specified in the contract or order, potassium nitrate shall be packed in accordance with best commercial practice, in textile-laminated-asphalt waterproof shipping sacks, drums or barrels. Drums or barrels of wood construction shall be lined with a waterproof, creped, bag-type liner. All containers shall conform to the requirements of Interstate Commerce Commission Regulations for Transportation of Explosives or Other Dangerous Articles, etc. so as to insure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

G–2. *Marking.*—

G–2a. *General.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

G–2b. *Army.*—Shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2.

G–2c. *Navy.*—Shipments for the Navy shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should specify the following:

Title, number and date of the specification
Class required (see pars. B–1 and E–1)
Type of packing required (see par. G–1).

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H–2. *Use.*—Potassium nitrate is intended for use as follows:
Class a—in black powder and chemical munitions
Class b—in pyrotechnic compositions
Class c—in propellants.

H–3. An explanation of the procedure for making a brass scale correction will be found in Lange's Handbook of Chemistry, page 1368, Fifth Edition (1944).

H–4. Copies of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. may be obtained upon application accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 40 cents.

H–5. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department". Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

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